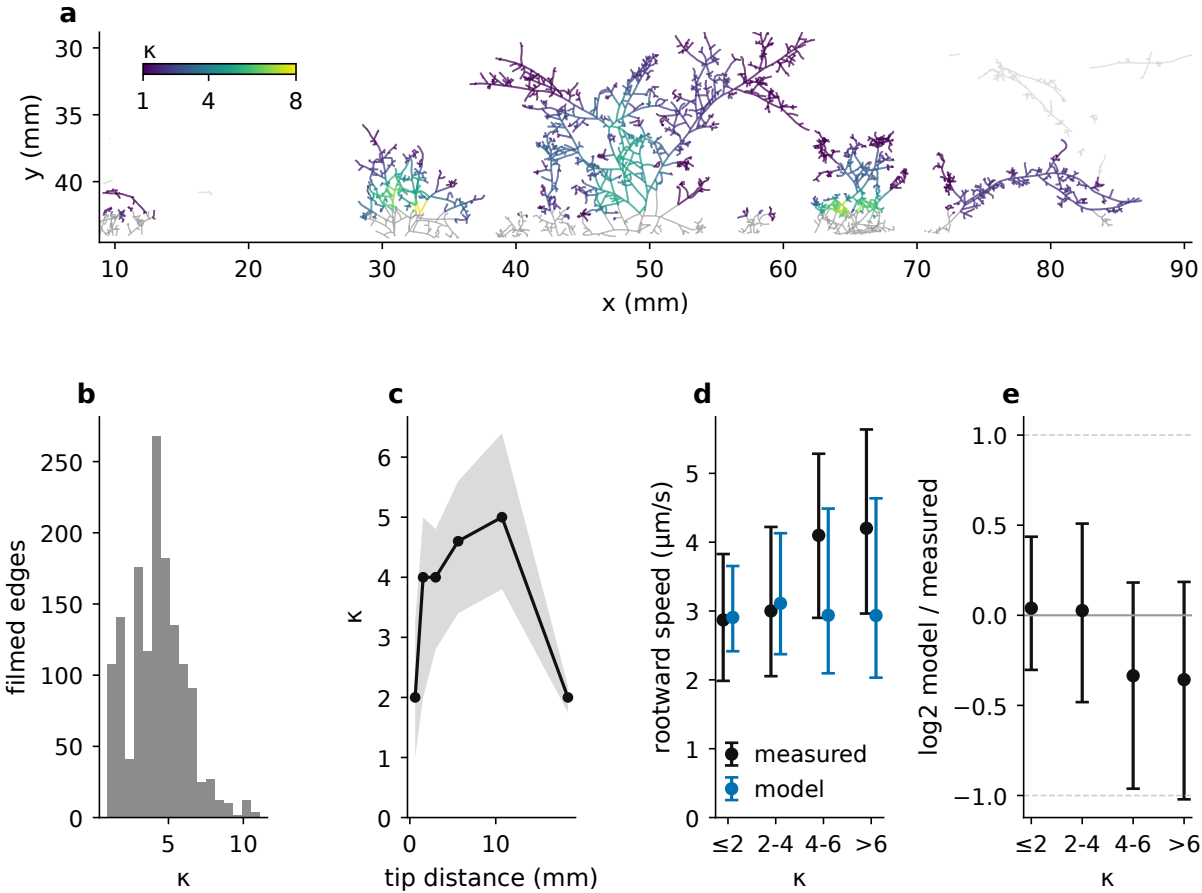

a
κ
1
4
8
y (mm)
30
35
40
10
20
30
40
50
60
70
80
90
x (mm)
b
filmed edges
250
200
150
100
50
0
5
10
κ
c
κ
6
5
4
3
2
1
0
0
10
tip distance (mm)
d
rootward speed (µm/s)
5
4
3
2
1
0
measured
model
≤2
2-4
4-6
>6
κ
e
log2 model / measured
1.0
0.5
0.0
−0.5
−1.0
≤2
2-4
4-6
>6
κ